Class XI Mathematics — Unit Test

Time: 90 minutes Maximum Marks: 25

SECTION A

1. Evaluate 2^3 + 1. 2

2. Solve 3x/4 + 2.5 = 7. 2

3. A force F = (3i + 4j) N acts on 2 kg. Find the acceleration. 3

4. The count is 6.02 x 10^{23}. Write it in standard form. 3

x 1 2 3

y 2 5 10

5. A projectile is fired with u = 20 m/s^2 and g = 9.8 m/s^2

at an angle of 30 degrees to the horizontal.

... Find the horizontal range of the projectile. 2

SECTION B

6. (a) Differentiate $x^3 + 2x$. 2

(b) Find the value of the derivative at $x = 2$. 3

7. (a) Solve $2x^2 - 5x + 2 = 0$. 2

(b) Find the discriminant $b^2 - 4ac$. 3

(c) Hence find the roots of the equation. 3